# A I CONCEPTS AND AI PROJECT FRAMEWORK

**What are the 3 classifications of AI?**

The 3 classifications of AI are

a) Weak AI
b) Strong AI and
c) Super AI

**What is week AI?**

Week AI is also known as narrow AI. It is the most common type of AI that we can see and use all around us . Narrow AI performs dedicated task with intelligence. It cannot perform beyond its capabilities .It is trained to perform specific task .E.g. Apple's Siri , IBM's Watson Supercomputer , Self- driven cars , Image processing ,e-commerce websites using chatbots, AI system playing chess ,etc.

**What is strong AI ?**

Strong AI is also termed as General AI or Broad AI .

Machines with the following characteristics will fall into this category.

Machines that would perform a wider range of different task with intelligence like that of humans

It is a step ahead of weak AI by including problem-solving , learning and development .

These systems are in their research stage.E.g.A machine hearing "Good Morning" and deciging to turn on the coffee maker.

**What is Super AI?**

Super AI s are in their hypothetical stage .It can be considered as a next stage of Strong AI and developing such machines will be a challenging task.

**What is an Expert System ?**

An expert system is a computer system that mimics the decision-making ability of a human who has expert knowledge and experience in a particular field e.g. ELIZA, MYCIN, DENDRAL

**Explain ELIZA**

The first Chabot ELIZA was created in MIT Artificial Intelligence Laboratory by Joseph Weizenbaum in 1966. It was a simulation of an Artificial Intelligence Rogerian psychotherapist.

**What is Computer Vision ?**

All machines use Computer Vision (CV) domain to understand their surroundings. Computer Vision is the domain of Artificial Intelligence that is used to train computers to read , process and analyse visual data in the same way as human do. E.g. A self driven car , facial lock in smart

phones, unusual behaviour detection, object classification, home security systems, office security systems and drone-based surveillance systems.

**What is Natural Language Processing (NLP) ?**

NLP gives the ability to machines to read, understand and analyse meaning from human language. The type of data used in NLP is text and speech. Applications of AI in NLP are are Chatbots , virtual assistants, auto-correct, sentiment analysis and auto-sentence completion.

**What are Neural Networks ?**

Neural networks are a series of algorithms used to recognise hidden patterns in raw data , cluster and classify it, and continuously learn and improve. They are used in a variety of applications in stock markets, sales and marketing trends, risk assessments and fraud detection. The main advantage is that the data features can be extracted automatically by the machine without input from the developer.

**Applications of Neural Networks**

a. Facial recognition – Cameras on smartphones can estimate the age of a person based on their facial features.
b. Forecasting- Neural networks are trained to understand the patterns and detect the possibility of rainfall or rise in stock prices with high accuracy.
c. Music composition- Neural networks can even learn patterns in music and train themselves enough to compose fresh tunes

**Advantages of Neural networks**

a. Parallel processing capability- Artificial Neural networks is a very powerful system that can perform more than one job at the same time
b. Data is stored on the entire network- Since the data is available on the entire network so if any node is down or unavailable , the whole system will not stop working
c. Capable of learning from non-linear and complex data- The input can be complex and non-linear for Artificial Neural networks to use to generate the desired output.

**Types of Neural Networks**

There are mainly two types of Neural Networks

a. Artificial Neural Networks (ANN)- It is an artificially created efficient computing system designed to simulate the human brain. It includes machine learning as a part of artificial intelligence. It is the foundation of AI and is used to solve complex problems that are difficult for humans. It consist of hardware and software that operates just like neurons of the human brain.

b. Biological Neural Networks- It is composed of a group of chemically connected or functionally associated neurons. Neurons transmit electrical signals to other neurons. These neurons are the building block of the complete central nervous system of the living body. The brain is the control unit of this neural network

# COMPONENTS AND STAGES OF AI PROJECT CYCLE

Project cycle is the steps taken to complete a task from the beginning to its end. There are mainly 5 stages of developing an AI Project

- Problem scoping
- Data acquisition
- Data exploration
- Modelling
- Evaluation

## 1. Problem scoping

**T**he first stage of an AI project cycle s problem scoping to identify the problem and to have a vision to solve it. Problems scoping means selecting a problem and finding a solution for it using AI technology.It focuses on identify and understanding problems using 4 Ws-Who, What, Where and Why .

**Understanding of problem** - The problem identifies the area in which AI can be used to provide a solution. Many times we are unable to observe any problem in our surroundings . In that case ,we can take a look at the sustainable development goals.

**Sustainable development goals**- When we cannot observe a problem around us then we should refer to the 17 goals that have been announced by the United Nations as the Sustainable development goals. These goals are to be achieved by 2030.

**Setting goals for an AI project**- Problem scoping is the term used to define the process of selecting a problem which we might want to solve using AI knowledge. Identifying a problem and then having a vision to solve it is called problem scoping.

**The 4 Ws Problem Canvas** -To understand step by step how problem scoping is done we use the method known as 4 Ws Problem Canvas . The 4 Ws are Who?, What?, Where? and Why? .

**Who?**- The “Who” block helps in analysing the people getting affected directly or indirectly due to it. Under this we find out who are the ‘Stakeholders’ to this problem and what we know about them . Stakeholders are the people who faces this problem and would be benefited with the solution.

**What?**- Under the “what” block, you need to look into what you have on hand . At this stage , you need to determine the nature of the problem. What is the problem and how do you know that it is a problem? Under this block , you also gather evidence to prove that the problem you

have selected actually exists. Newspaper, articles ,media, announcements ,etc are some examples .

**Where?**- Now that you know who is associated with the problem and what the problem actually is; you need to focus on the context /situation/ location of the problem . This block will help you look into the situation in which the problem arises, the context of it, and the locations where it is prominent.

**Why?** - You have finally listed down all the major elements that affect the problem directly. Now it is a convenient to understand who the people that would be benefited by the solution are; what is to be solved ; and where will the solution be deployed. These 3 canvases now become the base of why you want to solve this problem. Thus in the "Why" canvas think about the benefits which this stakeholders would get from the solution and how it will benefit them as well as his society.

**The Problem Statement**- After filling the 4Ws problem canvas , you need to summarise all the canvas into one template. The Problem Statement template helps us to put together all the key points into a single template for future reference.

**2. Data Acquisition**

This is the second stage of the AI project cycle. It is a process of collecting data required for training the AI project . Data is raw information that is used to generate meaningful outcomes

**Types of data**

There are 2 types of data

**Training data** - It is a data on which we train our AI project model . It is basically to fit the parameters of the project for the model. In training data , the output is available to the model

**Testing data** - It is used to check the performance of an A I model . In testing data , the data is not seen for which the predictions have to be made

**Data features-** Data features refers to the type of data that you want to collect

**Data sources** - Data is the base for any AI project to be built . When the data is acquired , it is important to check if it is from a reliable and authentic source for the accuracy of the project . There are various sources to collect relevant data for our project

- **Surveys** - Data can be collected from online surveys telephonic surveys or in- person surveys and collect responses. Surveys are a way of collecting data from a group of people in order to gain information and insights into various topics of interest.The process involves asking people for information through questionnaires which can be online or ofline. It can be considered as a data source

- **Web scraping-** Data or information can also be extracted from a website . Web scrapping or data scrapping is the method of downloading information from the world wide web

(WWW) and storing it on to your computer for later reference . The data collected in this way is an online data.

- **Sensors** - Data can also be collected from various sensors like collecting environmental data and storing it in some data storage solutions . Sensors are connected through gateways which enable them to collect live data in the ofline mode

- **Cameras-** Data can be seen , written down or recorded onto the computer. Cameras are used to collect data in the form of images, CCTV , web cameras , surveillance cameras are big sources of visual data that can be acquired from various places

- **Observation**- It is a method of collecting data by watching facts as they occur . Using the observation technique, data can be analysed and used for testing the model.

- **Application programming interface (APIs)** - APIs are a set of functions and procedures that allow one application to connect to another. So, one of the ways of collecting data is through APIs that can be used to collect data from social media services for analysis

**System Maps**

A system map is a diagrammatic representation of a set of things working together. It focuses on the components and boundaries of a system .System map helps us to find relationships between different elements of the problem which we have scoped .It helps to find a solution to achieve the goal of our project

**Importance of acquiring relevant data from reliable sources**

Acquiring relevant data from reliable sources is vital in AI as it ensures accurate and trustworthy model development

- Reliable data enhances the system's performance and generalisation capabilities
- It provides a solid foundation for making informed decisions and generating valuable insights
- Access to reliable data fosters ethical AI practices and promotes user trust

**Sources of Dataset in AI-Kaggle Platform**

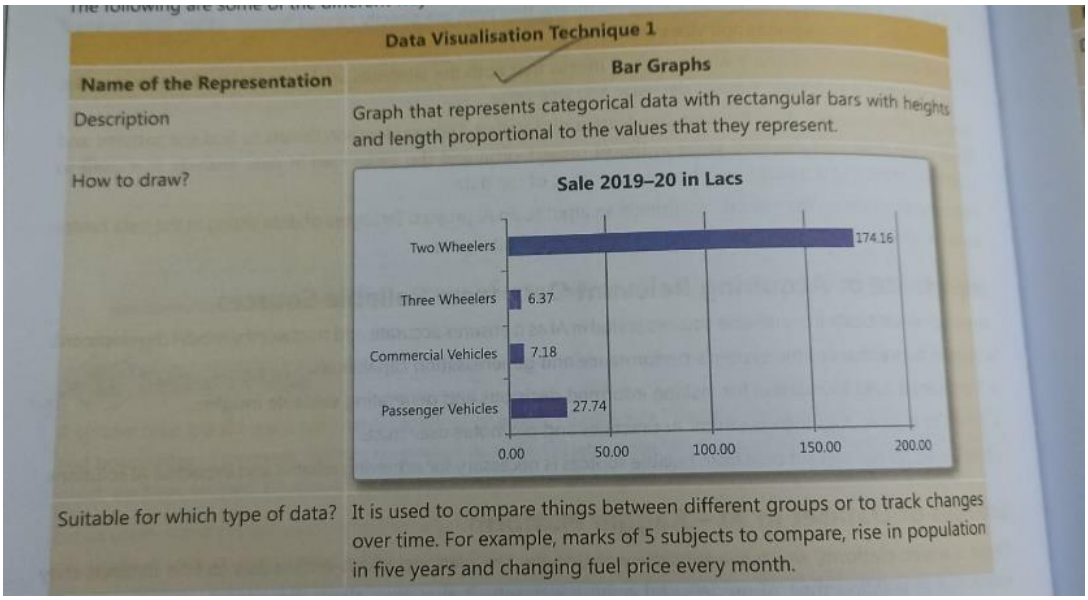
Kaggle is a web platform, which is a subsidiary of Google . The platform allows enthusiasts to find datasets they wish to use in building their AI models and publish datasets. It also gives them the opportunity to work with other data scientists and machine learning engineers. The Kaggle platform gives users the opinions of entering competitions to solve data science challenges.

3 Data exploration

Data exploration refers to exploring the large data to uncover the patterns or trends needed for the AI project. It is considered to be the first step in data analysis where unstructured data is explored , researched, filtered and visualised to decide this strategy for the type of model used in the later stage.

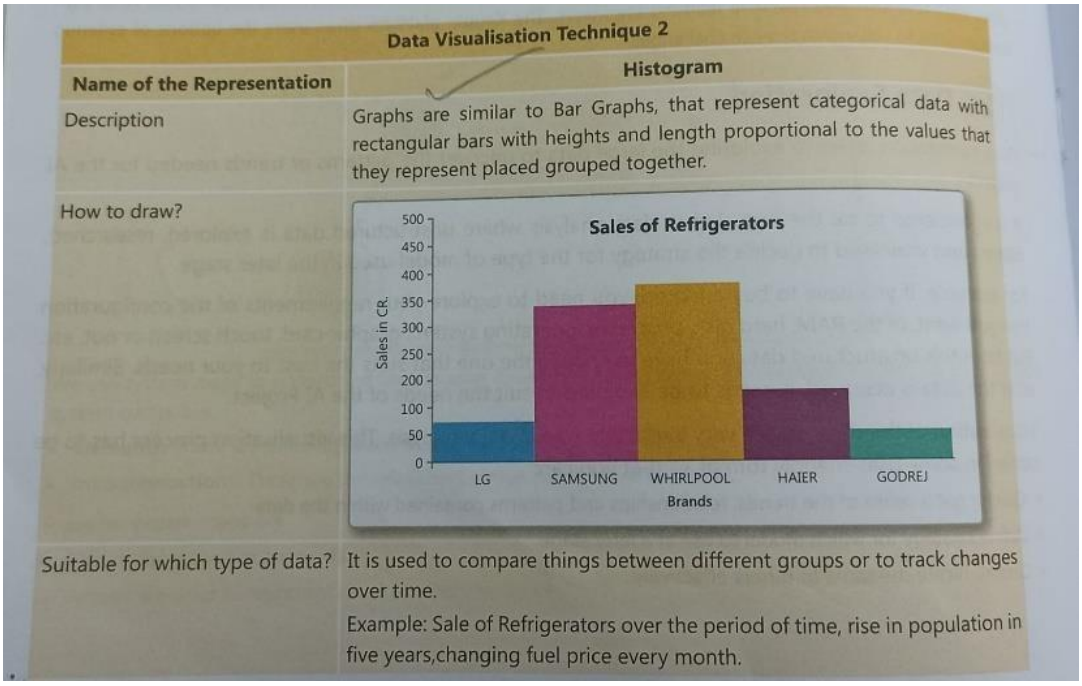
Different ways to visualise data

- Bar graphs



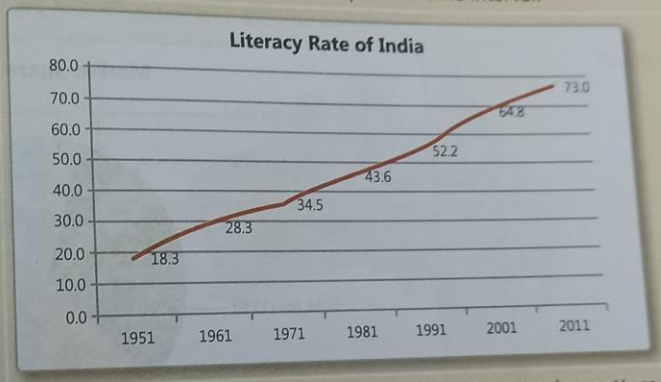
| Data Visualisation Technique 1 | |
|---|---|
| **Name of the Representation** | **Bar Graphs** |
| Description | Graph that represents categorical data with rectangular bars with heights and length proportional to the values that they represent. |
| How to draw? |  |
| Suitable for which type of data? | It is used to compare things between different groups or to track changes over time. For example, marks of 5 subjects to compare, rise in population in five years and changing fuel price every month. |

- Histogram



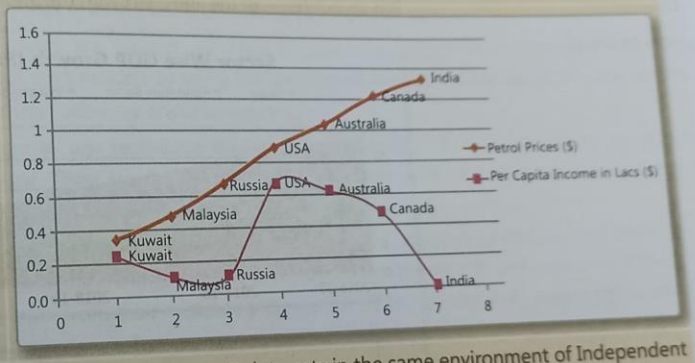
| Data Visualisation Technique 2 | |
|---|---|
| **Name of the Representation** | **Histogram** |
| Description | Graphs are similar to Bar Graphs, that represent categorical data with rectangular bars with heights and length proportional to the values that they represent placed grouped together. |
| How to draw? |  |
| Suitable for which type of data? | It is used to compare things between different groups or to track changes over time.<br>Example: Sale of Refrigerators over the period of time, rise in population in five years,changing fuel price every month. |

- Line graph

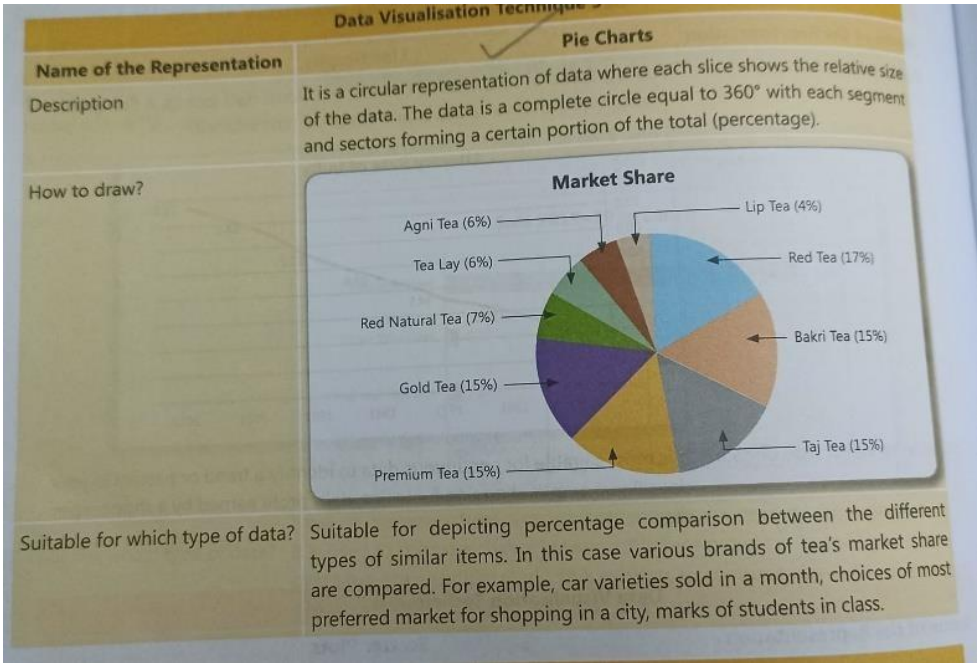
| Data Visualisation Technique 3 | |
|---|---|
| **Name of the Representation** | **Line Graphs** |
| Description | It is a graph that uses lines to connect individual data points. A line graph displays quantitative values over a specified time interval. |
| How to draw? |  |
| Suitable for which type of data? | It is most suitable for continuous data to identify a trend or pattern in your data. For example, changing fuel price, daily profit earned by a shopkeeper etc. |



- Scatter plot

| Data Visualisation Technique 4 | |
|---|---|
| **Name of the Representation** | **Scatter Plots** |
| Description | Graphs that present relationships between two variables in a dataset. Independent variables are plotted on the X-axis, while dependent variables are plotted on the Y-axis. |
| How to draw? |  |
| Suitable for which type of data? | It is used to compare 2 data sets in the same environment of Independent and Dependent Variables.<br>Example: rise in fuel prices in different countries over the period of time and per capita income in the same period, population size over the period of time in cities or countries and the disposable income during that period. |

- Pie Charts



| Name of the Representation | Pie Charts |
|---|---|
| Description | It is a circular representation of data where each slice shows the relative size of the data. The data is a complete circle equal to 360° with each segment and sectors forming a certain portion of the total (percentage). |
| How to draw? | Market Share: Agni Tea (6%), Lip Tea (4%), Tea Lay (6%), Red Tea (17%), Red Natural Tea (7%), Bakri Tea (15%), Gold Tea (15%), Taj Tea (15%), Premium Tea (15%) |
| Suitable for which type of data? | Suitable for depicting percentage comparison between the different types of similar items. In this case various brands of tea's market share are compared. For example, car varieties sold in a month, choices of most preferred market for shopping in a city, marks of students in class. |

**4 Modelling** - Modelling or data modelling is defined as a process of designing decision making algorithms that has to be trained on a set of data which was acquired at the data acquisition stage for the problem you scope indeed problem scoping stage and apply that learning to recognise certain types of patterns

**Approaches in AI modelling**- It is a mathematical approach in which an algorithm is designed as per the requirement of the system which is ready to be installed to analyse the data technically.

AI modelling techniques can be broadly classified into the following two approaches

- **Rule based approach** - This approach is based on a set of rules and facts defined by the developer and fed to the machine to perform its tasks accordingly to generate the desired output. These modls can operate with simple basic information and data.
- **Learning based approach** – This approach refers to the model where he relationship or patterns in the data are not defined by the developer. Random data is fed into the machine and the machine develops its own pattern or trends based on data outputs.

The learning based approach can be further divided into

- **Supervised learning** – In a Supervised learning model , a labelled data set is gven to the machine. A labelled data set is the information which is tagged with identifiers of data. For example clothes in a store are marked under various categories of clothing like Shirts, Trousers, Coats etc. They are further labelled as per gender ad size.
- **Unsupervised learning** – An Unsupervised learning approach works on an unlabelled data set. This means that the data which is fed to the machine is random and there is no know how available about it to the trainer.

Unsupervised learning can be further divided into

- **Clustering**- It is a machine learning approach where the machine partitions the data set into different clusters or categories based on machine generated algorithms. The data fed to such a moel is usually unlabelled or random and thus the developer feeds in the data directly into thee machine and instructs it to build its own algorithm.
- **Dimensionally Reduction**- Humans can visualise any figure upto three dimension only. But accordin to a lot theoris and algorithms, there are various entities which exists beyond three dimensions.

**Prediction** – Prediction in AI refers to the outcome/output of an algorithm after it has been trained on a historical dataset , used by the modelon the new dataset while forecasting the likelihood of a particular event / outcome.

**Linear Regression** – Regression is an example of rule-based AI model. Linear Regression is the most used regression modelling technique. It assumes that there is a linear relation between a dependent variable and independent variable which is reflected with a regression line and is a best fit line.

5 Evaluation – Evaluation is the last stage of the AI project cycle. It is an important step where the AI model is evaluated for its efficiency and accuracy . It enables continuous improvement of the model to achieve the project goals. The model must be tested with varied data to ensure that the results are satisfactory. Model is tested with the testing data after each stage of the AI project cycle.

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